

Sanjay Kumar Mucharla

Software Engineer

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SUMMARY

With over 4 years of experience, a Software Engineer specializing in developing scalable, high-performance web applications and enterprise solutions. Expertise spans backend development with Java, Python, Spring Boot, and Node.js, with a focus on building and maintaining RESTful APIs and microservices architectures. Proficient in creating responsive and interactive front-end interfaces using React.js, Redux, and UI frameworks such as Bootstrap5 and Tailwind CSS. Hands-on experience with cloud technologies, including AWS (Lambda, ECS, S3), Docker, and Kubernetes, ensuring efficient and reliable application deployments. Strong background in implementing real-time solutions, such as chat systems and data visualization, utilizing WebSockets, Redis, and GraphQL.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C, SQL, Go

Frontend Frameworks: React.js, Redux, Bootstrap5, Tailwind CSS, jQuery, HTML5, CSS3, AJAX, Angular, Vue.js

Backend Frameworks: Spring Boot, Node.js, Express.js, Django, Flask

Web Technologies: HTML5, CSS3, JSON, XML, AJAX, WebSockets, PWA (Progressive Web Apps)

Web Services: RESTful APIs, GraphQL, gRPC, SOAP

Databases: MySQL, PostgreSQL, MongoDB, Amazon RDS, DynamoDB, Firebase

Cloud & Application Servers: AWS (EC2, ECR, S3, Lambda, ECS, RDS, CloudFront, IAM), Docker, Kafka, Rancher, Kubernetes, Azure, Google Cloud Platform (GCP)

Version Control & CI/CD Tools: Git, GitHub, GitLab, Jenkins, CircleCI, Bitbucket, GitHub Actions

Testing Frameworks & Tools: JUnit, Mocha, Selenium, Enzyme, React Testing Library, Jest, Cypress, Postman

DevOps & Automation: Docker, Kubernetes, Jenkins, Terraform, Ansible, Prometheus, Grafana

Agile & Project Management: SDLC, Agile, Scrum, Kanban, Jira, Trello, Confluence

UI/UX Design Tools: Figma, Adobe XD, Sketch, InVision

Other Skills: Machine Learning, AI Integration, Data Structures, Algorithms, OAuth 2.0, JWT, OOP, Functional Programming, Microservices, Serverless Architecture

WORK EXPERIENCE

Software Engineer

Uber, FL

Jul 2024 – Present

- Developed a **fare optimization system** using **Java**, **Spring Boot**, and **MySQL**, reducing pricing discrepancies across regions by implementing real-time algorithms that process over 10 million daily transactions.
- Engineered a **ride-matching service** using **WebSockets**, **Node.js**, and **Redis**, enabling sub-5-second response times for trip allocations during high traffic.
- Automated deployment pipelines with **Jenkins**, **Docker**, and **Kubernetes**, reducing build and release times by 30 hours monthly.
- Designed and implemented a driver activity dashboard using React.js, Redux, and Tailwind CSS, translating business requirements into a responsive and interactive UI that improved driver engagement by 25%.
- Collaborated with cross-functional teams (DBAs, Infrastructure, Business) to integrate backend APIs with the frontend, ensuring seamless data flow and reducing latency by 30%.
- Automated deployment pipelines using Jenkins and Docker, reducing build and release times by 30 hours monthly and adhering to DevOps best practices.

Graduate Teaching Assistant

Florida International University, Miami, FL

Apr 2023 – Apr 2024

- Guided 50+ graduate students on building **RESTful microservices** using **Python**, **Flask**, and **MongoDB**, ensuring comprehensive project completion.
- Built a location-based property search module using TypeScript, React.js, and GraphQL, handling 100,000+ daily queries with a 30% improvement in search latency.
- Translated business requirements into technical solutions by working closely with the business team, ensuring alignment with project goals and delivering scalable, maintainable code.
- Developed and maintained RESTful APIs using Node.js and PostgreSQL, achieving 99.9% uptime for critical services and ensuring seamless integration with the frontend.

Software Developer

Airbnb, India

Feb 2021 – Jul 2022

- Built a **location-based property search module** leveraging **TypeScript**, **React.js**, and **GraphQL**, which supported over 100,000 queries daily.
- Enhanced booking reliability by developing a **high-availability payment gateway** using **Spring Boot** and **PostgreSQL**, achieving uptime.
- Automated property listing validation processes with **Selenium** and **JUnit**, reducing manual intervention time by 40 hours monthly.
- Architected and containerized a microservices-based notification system using **Docker** and **AWS ECS**, ensuring seamless delivery of 500,000 daily alerts.
- Designed **real-time chat functionality** between hosts and guests with **Node.js**, **WebSockets**, and **MongoDB**, enhancing user satisfaction scores.
- Integrated **CI/CD pipelines** using **Jenkins** and optimized deployment cycles, reducing production errors in staging environments.

Software Developer Intern

Trigent Software, India

Jul 2020 – Jan 2021

- Assisted in developing an **inventory management system** for retail clients using **Java**, **Spring MVC**, and **MySQL**, enabling accurate tracking of over 1 million SKUs.
- Implemented **AJAX-based filtering** on inventory search pages, improving user experience and reducing query latency by 4 seconds.
- Created and integrated **interactive dashboards** with **React.js** and **Chart.js** to visualize sales and product trends in real time.
- Streamlined testing workflows by automating UI testing with **Selenium WebDriver**, reducing regression testing time by 15 hours per sprint.
- Enhanced application security by implementing **OAuth 2.0** authentication for RESTful APIs, minimizing security incidents.
- Assisted in developing an inventory management system using Java, Spring MVC, and MySQL, enabling accurate tracking of over 1 million SKUs and improving operational efficiency.
- Implemented AJAX-based filtering on inventory search pages, reducing query latency by 4 seconds and enhancing user experience.
- Conducted thorough testing and debugging in a testing environment, identifying and resolving errors to ensure high-quality deliverables.

PROJECT

Smart Web Fill

Jun 2023 – Dec 2023

Technologies: JavaScript, HTML, CSS, .NET API, Azure SQL Database, Microsoft AI Form Recognizer

- Built an **AI-driven Smart Form Fill application** with **OCR** and **Microsoft AI Form Recognizer**, enabling automatic extraction and population of web forms from uploaded documents, reducing manual entry errors by 70%.
- Developed a **RESTful API** backend in **.NET** and an interactive frontend with **JavaScript** and **React.js**, facilitating seamless integration with **GitHub OAuth** for enhanced collaboration data visualization.

Real-Time Chat Application

Jan 2024 – Mar 2024

Technologies: WebSocket, Node.js, Redis, Machine Learning

- Implemented a **real-time chat platform** using **Node.js** and **WebSocket**, capable of handling over 10,000 concurrent users while ensuring end-to-end encrypted communication.
- Integrated **ML-based real-time language translation**, enabling users to communicate across 5+ languages seamlessly, enhancing accessibility and engagement.

EDUCATION

Florida International University

Miami, FL

Master of Computer Science

Aug 2022 – Apr 2024

Bharath University

Chennai, India

Bachelor's in Computer Science

Aug 2017 – May 2021